

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____M.NITHIYASHREE-192421101_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____M.NITHIYAHSREE_____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud.
Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems.

Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.

Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment.

Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage

Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWER

1. Company with Seasonal Sales Traffic Spikes

Application Scenario

An e-commerce company experiences a sudden increase in website traffic during seasonal sales such as Black Friday or Diwali.

Cloud Auto-Scaling and Load Balancing

- **Auto-scaling** automatically increases the number of application servers when traffic increases and removes extra servers when demand decreases.
- **Load balancing** distributes incoming user requests evenly across all available servers, preventing any single server from becoming overloaded.

Reducing Downtime and Improving User Experience

- Prevents server crashes during peak traffic.
- Reduces website response time.
- Ensures continuous availability of the online store.
- Improves customer satisfaction with faster page loading and checkout.

Role of Monitoring Tools

Cloud monitoring services (such as Amazon CloudWatch, Azure Monitor, or Google Cloud Monitoring) continuously monitor:

- CPU usage

- Memory usage
- Network traffic
- Response time
- Error rates

These tools automatically trigger auto-scaling policies and send alerts if any issues occur.

Real-World Example

During **Amazon Prime Day**, millions of users access the website simultaneously. Auto-scaling adds more servers, while load balancers distribute traffic efficiently, ensuring smooth shopping without downtime.

2. Secure Cloud Backup Storage

Application Scenario

An organization stores large volumes of backup data in the cloud for disaster recovery.

Durability and Data Redundancy

Cloud storage providers improve reliability by:

- Storing multiple copies of data across different servers.
- Replicating data across multiple Availability Zones or regions.
- Automatically repairing corrupted or lost data.

Encryption and Access Control

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.

Access Control

- Use Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).
- Apply Identity and Access Management (IAM) policies.

Cost Advantages

- No need to purchase physical storage hardware.
- Pay only for the storage used.
- Reduced maintenance and electricity costs.
- Automatic scaling without additional infrastructure.

Disaster Recovery Example

If a company's primary data center is damaged due to flooding, backup data stored in the cloud can be restored quickly from another region, minimizing downtime and data loss.

3. Global Mobile Application Deployment

Application Scenario

A mobile application company wants to provide fast access to users worldwide.

Role of Content Delivery Network (CDN)

A CDN stores copies of application content on edge servers located in different geographical regions.

When users request content, it is delivered from the nearest edge server instead of the central server.

Benefits of Edge Servers

- Lower network latency.
- Faster content delivery.
- Reduced bandwidth usage.
- Better application performance.
- Improved user experience.

Cloud Provider Global Availability

Cloud providers maintain:

- Multiple global data centers.
- Regional Availability Zones.
- Global load balancers.
- Automatic traffic routing.

This ensures continuous service availability even during regional failures.

Real-World Example

Netflix uses CDNs to stream movies from nearby edge servers, reducing buffering and providing smooth video playback worldwide.

4. Financial Institution Cloud Security and Compliance

Application Scenario

A bank stores customer financial records in the cloud and must comply with financial regulations.

Regulatory Compliance

Cloud providers support standards such as:

- ISO 27001
- PCI DSS
- GDPR
- SOC 2

These standards ensure secure handling of financial information.

Role of Logging, Monitoring, and Encryption

Logging

- Records all user activities and system events.
- Supports auditing and investigations.

Monitoring

- Detects suspicious activities.
- Sends alerts for security incidents.
- Monitors system health.

Encryption

- Protects sensitive customer information.
- Encrypts stored and transmitted data.

Data Sovereignty

To meet legal requirements, organizations:

- Store data within approved geographic regions.
- Select cloud regions that comply with national regulations.
- Control where customer data is processed.

Example

When customers perform online banking transactions, all communication is encrypted using TLS. Every transaction is logged for auditing, while monitoring tools detect unauthorized access attempts and suspicious activities.

5. Containerization for Application Deployment

Application Scenario

A software development team plans to deploy applications using containers.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Performance	High	Moderate
Operating System	Shares host OS	Separate guest OS
Portability	High	Moderate

Kubernetes Orchestration

Kubernetes simplifies deployment by:

- Automatically deploying containers.
- Scaling applications based on demand.
- Replacing failed containers automatically.
- Managing service discovery.
- Balancing application traffic.

Portability Benefits

Containers package the application with all required libraries and dependencies.

This allows the same container to run consistently on:

- AWS
- Microsoft Azure
- Google Cloud Platform
- Private cloud
- On-premises servers

Microservices Example

An **online shopping application** can be divided into independent services:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Management Service
- Notification Service

Each microservice runs inside its own container and can be deployed, updated, and scaled independently.

Conclusion

Cloud technologies improve scalability, security, and performance for modern applications. **Auto-scaling and load balancing** handle traffic spikes efficiently, while **cloud storage** provides secure and durable backups. **CDNs and edge servers** deliver content quickly to global users. **Compliance, encryption, logging, and monitoring** ensure secure cloud operations for sensitive industries like finance. **Containers and Kubernetes** enable efficient deployment, portability, and scalability, making them ideal for modern microservices-based applications.